

LLM & GENAI

- vLLM
- Unsloth
- LoRA / QLoRA
- PEFT
- Transformers
- Self-hosted LLM Serving
- LLM Evaluation

ML & DEEP LEARNING

- XGBoost
- Autoencoders
- PyTorch
- TensorFlow
- Keras
- DenseNet
- Logistic Regression
- Computer Vision
- Time-Series Forecasting
- Behavioral Anomaly Detection
- CUDA GPU Libraries
- RAPIDS cuDF / CuPy
- Optimisation & Simulation
- Google OR-Tools
- Vehicle Routing
- Mixed-Integer Programming
- Operations Research
- Constraint Modelling
- Supply-Chain Optimization

MLOPS & PLATFORMS

Cloud & Orchestration

- GCP Vertex AI
- GKE
- Dataflow / Apache Beam
- Pub/Sub
- Azure ML Studio
- Azure DevOps

Serving & Packaging

- Kubernetes
- Helm
- Istio
- Docker
- Seldon Core
- Cloud Build
- ArgoCD
- Google KCC

Observability & CI/CD

- Grafana
- Prometheus
- Model Monitoring
- CI/CD
- GitOps
- Autoscaling
- Blue-Green Delivery
- Load Testing

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer

Best Buy · Bengaluru, India

Jul 2025 – Present

 - Built a configurable **LoRA fine-tuning framework** using Unsloth, PEFT, and QLoRA; supported single- and multi-GPU runs and multi-task single-adaptor training.
 - Deployed OpenAI-compatible vLLM serving on GKE with HPA autoscaling and Istio routing; integrated GCS storage, Artifact Registry versioning, Cloud Build CI, and Prometheus/Grafana monitoring and alerting. Self-hosting eliminated paid API dependency, with estimated annual API-cost savings of **~\$300K**, and strengthened data governance and compliance posture. Kept model inputs company-managed and reduced external API exposure during peak demand.
 - Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.
- Lead ML Engineer — ML Engineering & Operations

Wipro · Client: Major US Retail Corporation

Aug 2023 – Jul 2025

 - Built AWS account anomaly detection using per-entity autoencoders and NVIDIA Morpheus on **DGX (8×A100)**. Trained on 20+ days of logs across multiple AWS accounts, totaling **billion-row datasets**, to detect deviations from normal activity in real time.
 - Built NVIDIA Morpheus phishing detection processing **2,400 emails/sec**, doubling throughput; automated abuse-inbox triage by extracting scammer identifiers and routing threats to security teams.
 - Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; completed **300+ résumé screens** and conducted **50+ technical interviews**.
- Senior Data Scientist — Data Science / Azure ML

Accenture · promoted from Senior Analyst

May 2021 – Sep 2023

 - Developed an Azure ML pipeline for BamaGruppen (Norway), delivering **\$2.5M annual savings**; increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**.
 - Led last-mile delivery optimization with OR-Tools, improving route planning, departure scheduling, and load distribution under vehicle-capacity and time-window constraints.
 - Designed supply-chain optimization for hub-location and distribution decisions; led cross-functional delivery, mentored junior data scientists, and presented recommendations to Norwegian stakeholders.

LANGUAGES & DATA

Python SQL Pandas
MS SQL Server Power BI

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

Microsoft Certified
Power BI Data Analyst Associate

EDUCATION

B.E. Electrical & Electronics

Andhra University, Visakhapatnam
2012 – 2016 · 80%

Achievements

15 honors and awards across Best Buy, Wipro, Accenture, and Infosys. Best Buy client AML Hall of Fame (2024): DGX delivery during the Wipro engagement. Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.

GPU & DISTRIBUTED COMPUTING

GPU-Accelerated Computing

MODEL LIFECYCLE

Experiment Tracking
Model Versioning / Registry
Automated Retraining
Vertex AI Pipelines
Azure ML Pipelines
Vertex AI Feature Store

PRODUCTION ENGINEERING

ML System Architecture
Cloud-Native ML Systems
JWT Authentication
Unit / Integration Testing

ENGINEERING LEADERSHIP

Architecture Reviews
Technical Mentoring
Technical Hiring Team Building
Stakeholder Alignment
Cross-Functional Delivery

EARLIER EXPERIENCE

Data Scientist

GD Research Centre Pvt Ltd
May 2020 – May 2021

Built company-wide time-series forecasting across multiple intelligence centers (Consumer Goods, Oil & Gas, Power, Retail, Finance).

Deep Learning Engineer

Tata Consultancy Services
Jul 2019 – Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on ~1 million images in ~3 hours; 90% classification accuracy. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer — ML

Infosys
May 2016 – Jul 2019

Built an ML-powered lead-conversion prediction system and business intelligence applications using Microsoft BI tools.

PROJECTS

LLM Fine-tuning & Serving Framework

Best Buy · Jul 2025 – Present

Config-driven LoRA adapter training for reusable tasks and open-source causal LLMs. The framework supports Unsloth/PEFT and QLoRA across single- and multi-GPU runs. vLLM serves adapters on GKE with HPA autoscaling, Istio routing, and GCS-backed model storage.

QLoRA GCS

~\$300K estimated annual API-cost savings

AWS Account Behavioral Fingerprinting

Wipro · Aug 2023 – Jul 2025

Unsupervised autoencoder models trained per-entity (account / user / service / machine) to profile normal AWS activity patterns and surface anomalous deviations in real time. Training and inference pipelines engineered with NVIDIA Morpheus on DGX (8xA100). 20+ days of logs across multiple AWS accounts totaling billion-row datasets; model management via MLflow; alerts served via Grafana.

RAPIDS / cuDF MLflow

Billion-row datasets · Per-entity autoencoders · 8xA100 DGX

Last-Mile Delivery Optimization

Accenture · Aug 2022 – Sep 2023

Vehicle Routing Problem (VRP) formulation using Google OR-Tools for a major logistics operator — optimizing route planning, departure scheduling, and product load distribution across a delivery fleet. Constraint modelling accounted for time windows, vehicle capacity, and geographic clustering.

OR-Tools VRP Simulation

Automated daily transport planning · OR-Tools VRP at fleet scale

Product Quality Prediction — BamaGruppen

Accenture · May 2021 – Dec 2021

End-to-end Azure ML pipeline predicting product quality at warehouse arrival for a Norwegian retail group — enabling proactive inspection scheduling and reducing downstream quality failures. Fully operationalized with automated triggers and pipeline orchestration.

Logic Apps Azure Pipelines Azure Storage

\$2.5M/year · 130 → 160 trucks/day · +17% detection · +20% approval

Supply-Chain Optimization & Hub Planning

Accenture · Nov 2022 – Jul 2023

Built a simulation and optimization framework covering vendors, warehouses, distribution hubs, and customers. Compared operating cost, product freshness, and network scenarios to identify the appropriate number and locations of regional hubs.

Azure ML

Informed regional hub-location and distribution decisions

Industry Forecasting Platform

GD Research Centre · May 2020 – May 2021

Designed domain-aware time-series forecasting and scenario analysis for GlobalData UK across energy, manufacturing, healthcare, finance, retail, telecommunications, automotive, and other industry intelligence centers.

Time Series Feature Engineering Scenario Analysis

15+ industry verticals · Enterprise market-intelligence forecasts